

Abdelrahman Khalifa

Software Engineer — C++ · Rust · Linux · Codeforces Expert

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SUMMARY

Software engineer with over a year of professional C++ work, a completed Google Summer of Code 2026 project in Rust (The Linux Foundation) and 7 PRs merged upstream. Codeforces Expert and 4-time ECPC finalist; at home in large, unfamiliar codebases and strict code review.

TECHNICAL SKILLS

Languages: C++ (modern C++, Qt), Rust, C, Python, Java, C#, SQL, JavaScript, Bash

Computer science: Data structures and algorithms, operating systems, concurrency and multithreading, async/event-driven design, memory management, networking (TCP/UDP, IPC)

Systems & Linux: Linux (Ubuntu/Debian), FFI, IPC (varlink, D-Bus), CUPS/IPP, DNS-SD/mDNS, kernel modules and device drivers, Yocto, Buildroot, QEMU

Tools & web: Git/GitHub, CMake, Make, Cargo, CI, testing (unit, end-to-end, pytest), debugging, code review; React, Node.js, MongoDB, REST APIs

OPEN SOURCE

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC desktop printer setup tool in Rust | github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers

- Designed and built a printer-management stack for the COSMIC desktop as 5 Rust crates: shared types, an async client, a CUPS/IPP/DNS-SD backend, a shared UI and a standalone app.
- Isolated the C FFI backend behind a varlink IPC service in the settings daemon, so a crash in native code cannot take down the Settings app.
- 7 PRs merged upstream:** ported cups-rs to libcups3 with CUPS 2 and CUPS 3 builds and CI, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).
- Tested on real HP printers; CI runs formatting, Clippy and tests on several architectures, with virtual IPP printers.

Contributor — Boa JavaScript engine (Rust), Rust Clippy, cups-rs, Zephyr RTOS | Mar 2024 – Jan 2026

- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion; with tests ([PR #4553](#)).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` ([PR #16214](#)).
- Winter of Code 2026 (OpenPrinting): implemented the `lp` and `lpstat` commands in Rust on cups-rs ([PR #11](#)).
- Zephyr RTOS: implemented the `IP_MULTICAST_LOOP` socket option across 6 networking modules, merged upstream ([commit b117036](#)).

EXPERIENCE

Embedded Systems Engineer — EKSON | Aug 2025 – Present

- Built a modern C++/Qt desktop control application for motion simulators and optimized its performance by ~40% through asynchronous processing and fewer threads; reliable UART control.
- Developed Zephyr RTOS firmware for an ESP32-S3 3-DoF simulator, migrating from FreeRTOS for more deterministic behavior; integrated an MPU6050 IMU with calibration and filtering.

Embedded Linux Systems Intern — ODC | Dec 2023 – Jan 2024

- Built Linux systems with Yocto and Buildroot; worked with kernel modules, device drivers, device trees, file systems and hardware flashing.

PROJECTS

- Operating System Simulator (Java):** CLI that simulates process scheduling, memory management, system calls and locks. [GitHub](#)
- IoT Device Communication System (C++, Raspberry Pi 4, QEMU, Yocto):** object-oriented framework for TCP unicast and UDP multicast between a server and emulated clients. [GitHub](#)
- Inventory Management System (C#, WinForms, SQL):** desktop app for stock, orders and deliveries with role-based access control and automated reports. [GitHub](#)
- Team projects (co-developed):** Sarh Al-Tafawuq, a production web platform ([live](#)); Nafaqati, a Flutter Android expense tracker with on-device SMS import and offline storage ([Google Play](#)).

ACHIEVEMENTS

- Codeforces Expert (handle `_Khalifa`) and 4-time ECPC finalist (Egyptian Collegiate Programming Contest).
- Deloitte Software Engineering Mentorship (React Native, 2024); IEEE student chapter Treasurer (\$10k+ budget).

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 – 2025

Very Good (84.6%) with Honors. Graduation project: AUTOSAR-based V2X vehicle prototype.